

# Analyzing Traffic with Wazuh

**Project Overview:** This project's main purpose is to architect a local security monitoring environment to simulate, detect, and analyze network-based attacks. The lab environment consisted of an Ubuntu server virtual machine hosting the Wazuh SIEM manager, a Windows virtual machine serving as the target (configured with the Wazuh agent and Sysmon for advanced endpoint telemetry), and a Kali Linux virtual machine acting as the attack machine. The Kali machine utilized a variety of different tools for both reconnaissance such as Nmap, as well as hydra for the actual brute-force attack. The project demonstrated in the engineering of custom Wazuh detection rules designed to successfully identify and alert on the automated credential stuffing attempts, demonstrating a practical understanding of both offensive tactics and defensive security operations.

**Objectives:** Connect windows virtual machine to the Wazuh manager, run a simulated attack in a controller environment using a second kali virtual machine on the same internal network, and investigate system notifications

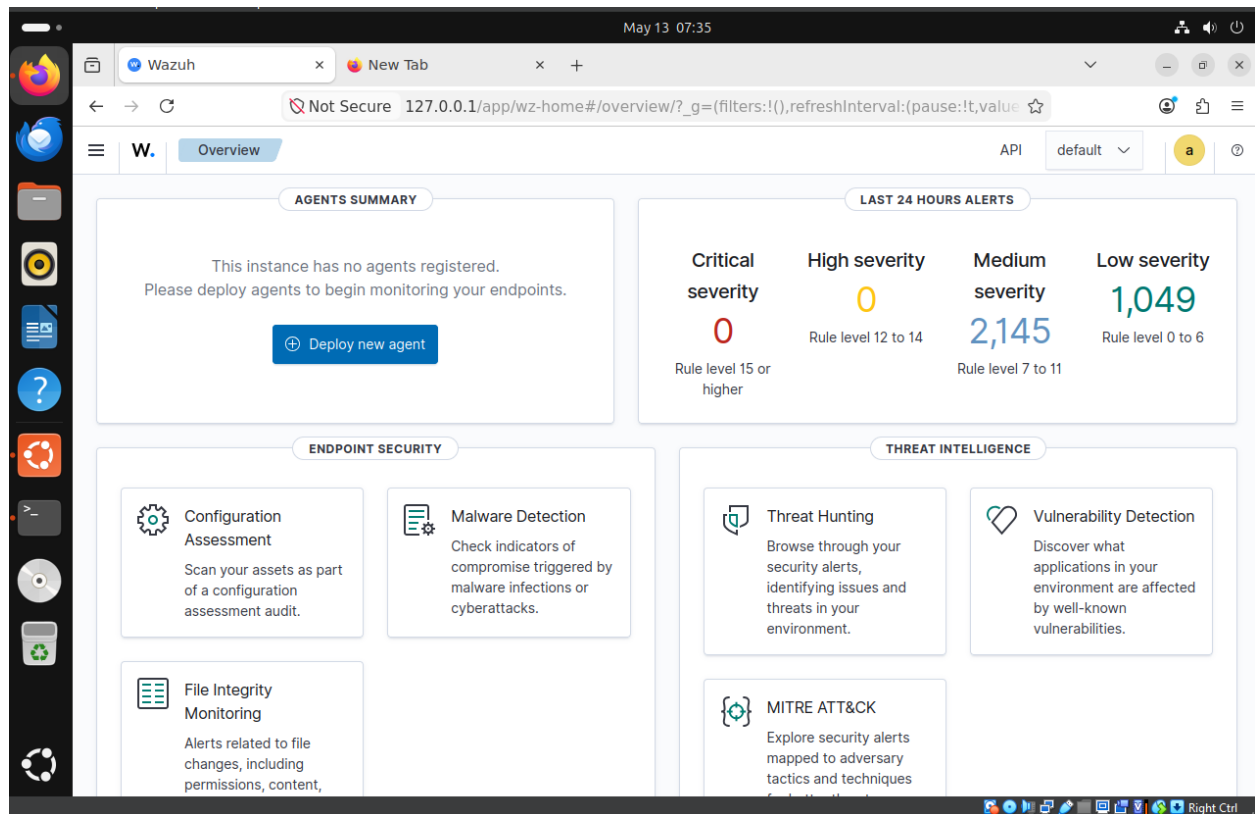
First, I used this command in the screenshot below to install the open source SIEM software called Wazuh on a Ubuntu Linux server virtual machine.

```

vboxuser@ubuntu1:~$ curl -sO https://packages.wazuh.com/4.14/wazuh-install.sh && sudo bash ./wazuh-install.sh -a
13/05/2026 07:02:32 INFO: Starting Wazuh installation assistant. Wazuh version: 4.14.5
13/05/2026 07:02:32 INFO: Verbose logging redirected to /var/log/wazuh-install.log
13/05/2026 07:02:32 INFO: The recommended systems are: Red Hat Enterprise Linux 7, 8, 9; CentOS 7, 8; Amazon Linux 2; Amazon Linux 2023; Ubuntu 16.04, 18.04,
20.04, 22.04; Rocky Linux 9.4.
13/05/2026 07:02:32 WARNING: The current system does not match with the list of recommended systems. The installation may not work properly.
13/05/2026 07:02:35 INFO: Verifying that your system meets the recommended minimum hardware requirements.
13/05/2026 07:02:35 INFO: Wazuh web interface port will be 443.
13/05/2026 07:02:39 INFO: --- Dependencies ---
13/05/2026 07:02:39 INFO: Installing apt-transport-https.
13/05/2026 07:02:40 INFO: Installing debhelper.
13/05/2026 07:02:50 INFO: Wazuh repository added.
13/05/2026 07:02:51 INFO: --- Configuration files ---
13/05/2026 07:02:51 INFO: Generating configuration files.
13/05/2026 07:02:51 INFO: Generating the root certificate.
13/05/2026 07:02:51 INFO: Generating Admin certificates.
13/05/2026 07:02:51 INFO: Generating Wazuh indexer certificates.
13/05/2026 07:02:51 INFO: Generating Filebeat certificates.
13/05/2026 07:02:51 INFO: Generating Wazuh dashboard certificates.
13/05/2026 07:02:51 INFO: Created wazuh-install-files.tar. It contains the Wazuh cluster key, certificates, and passwords necessary for installation.
13/05/2026 07:02:52 INFO: --- Wazuh indexer ---
13/05/2026 07:02:52 INFO: Starting Wazuh indexer installation.
13/05/2026 07:03:19 INFO: Wazuh indexer installation finished.
13/05/2026 07:03:19 INFO: Wazuh indexer post-install configuration finished.
13/05/2026 07:03:19 INFO: Starting service wazuh-indexer.
13/05/2026 07:03:30 INFO: wazuh-indexer service started.
13/05/2026 07:03:30 INFO: Initializing Wazuh indexer cluster security settings.
13/05/2026 07:03:32 INFO: Wazuh indexer cluster security configuration initialized.
13/05/2026 07:03:32 INFO: Wazuh indexer cluster initialized.
13/05/2026 07:03:32 INFO: --- Wazuh server ---
13/05/2026 07:03:32 INFO: Starting the Wazuh manager installation.
13/05/2026 07:05:28 INFO: Wazuh manager installation finished.
13/05/2026 07:05:28 INFO: Wazuh manager vulnerability detection configuration finished.
13/05/2026 07:05:28 INFO: Starting service wazuh-manager.
13/05/2026 07:05:41 INFO: wazuh-manager service started.
13/05/2026 07:05:41 INFO: Starting Filebeat installation.
13/05/2026 07:05:46 INFO: Filebeat installation finished.
13/05/2026 07:05:47 INFO: Filebeat post-install configuration finished.
13/05/2026 07:05:47 INFO: Starting service filebeat.
13/05/2026 07:05:48 INFO: filebeat service started.
13/05/2026 07:05:48 INFO: --- Wazuh dashboard ---
13/05/2026 07:05:48 INFO: Starting Wazuh dashboard installation.
13/05/2026 07:06:50 INFO: Wazuh dashboard installation finished.
13/05/2026 07:06:50 INFO: Wazuh dashboard post-install configuration finished.
13/05/2026 07:06:50 INFO: Starting service wazuh-dashboard.
13/05/2026 07:06:50 INFO: wazuh-dashboard service started.
13/05/2026 07:06:59 INFO: Updating the internal users.
13/05/2026 07:07:02 INFO: A backup of the internal users has been saved in the /etc/wazuh-indexer/internalusers-backup folder.
13/05/2026 07:07:09 INFO: The filebeat.yml file has been updated to use the Filebeat Keystore username and password.
13/05/2026 07:07:35 INFO: Initializing Wazuh dashboard web application.
13/05/2026 07:07:36 INFO: Wazuh dashboard web application initialized.

```

After this was done I installed a Ubuntu GUI on the server for easier management of the Wazuh dashboard using the command “sudo apt update && sudo apt upgrade -y” to update system packages.” sudo apt install tasksel -y” to install the software tasksel for the actual GUI, and then use “sudo tasksel” to run the software and then selected Ubuntu desktop to select which interface to use, then reboot the server for it to take effect, We then log into the GUI portal for Wazuh.



Then installed Wazuh agent on a separate windows VM, after many trial and error of the authentication key not importing to the agent machine because of a human error of using the wrong IP address for the manager server.

Select Administrator: Windows PowerShell

Windows PowerShell

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Try the new cross-platform PowerShell <https://aka.ms/pscore6>

```
PS C:\Windows\system32> Invoke-WebRequest -Uri https://packages.wazuh.com/4.x/windows/wazuh-agent-4.14.5-1.msi -OutFile $env:tmp\wazuh-agent; msixec.exe /i $env:tmp\wazuh-agent /q WAZUH_MANAGER='127.0.0.1'
```

```
PS C:\Windows\system32> NET START Wazuh
```

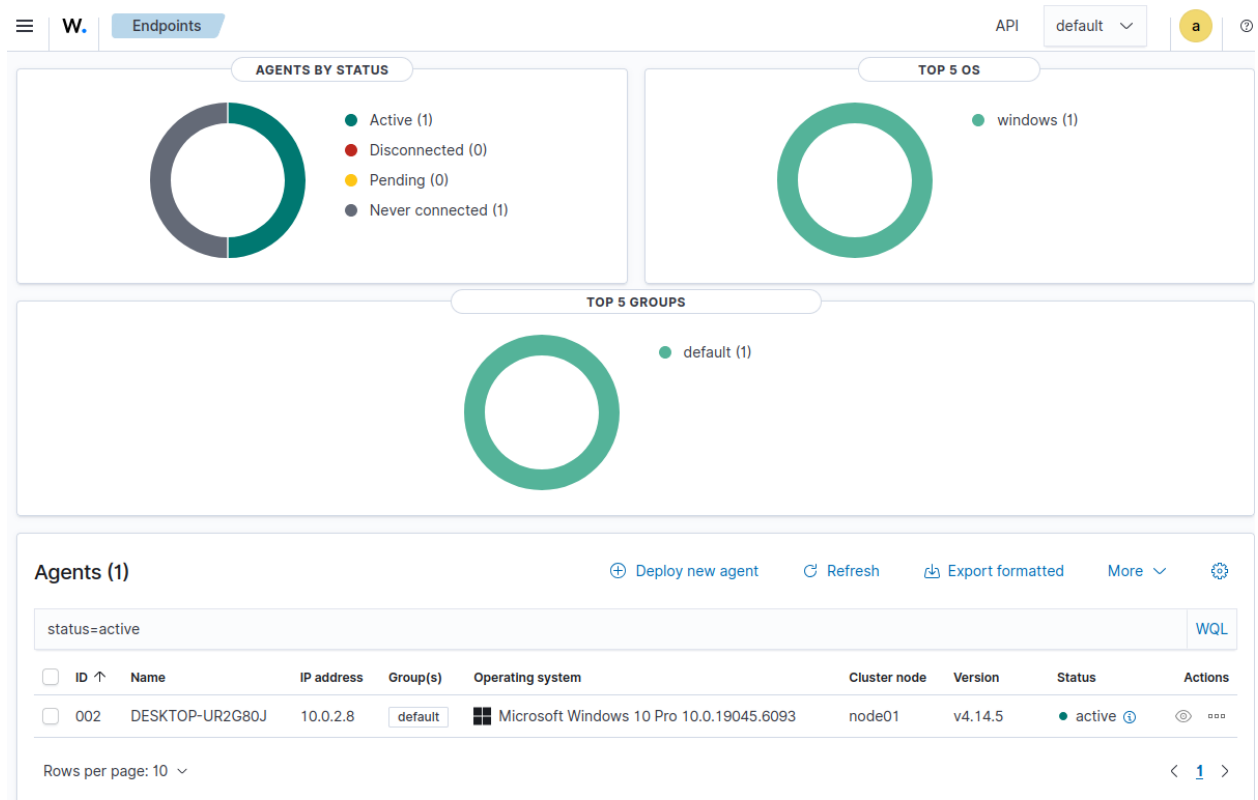
The Wazuh service was started successfully.

```
PS C:\Windows\system32> Invoke-WebRequest -Uri https://packages.wazuh.com/4.x/windows/wazuh-agent-4.14.5-1.msi -OutFile $env:tmp\wazuh-agent; msixec.exe /i $env:tmp\wazuh-agent /q WAZUH_MANAGER='10.0.2.10'
```

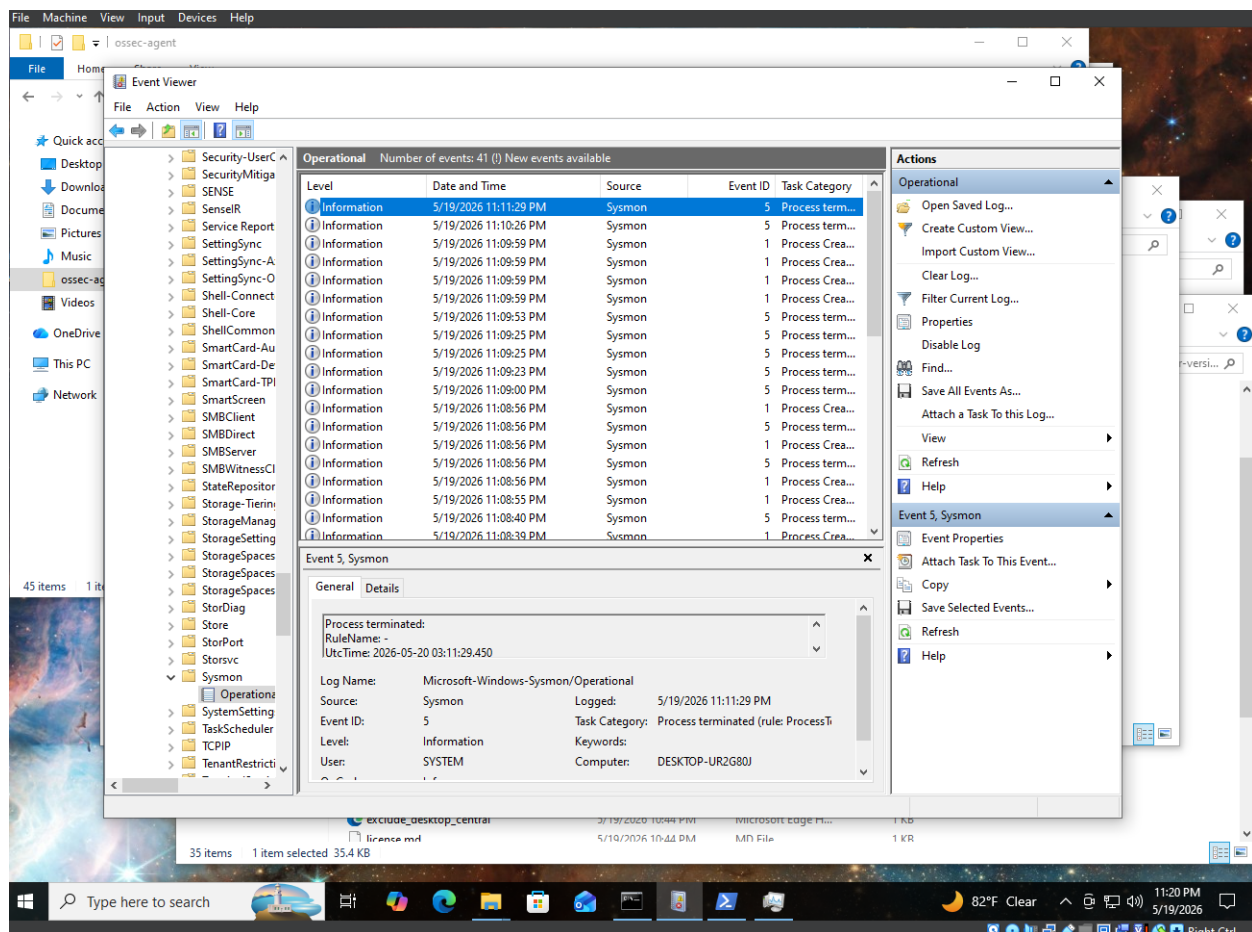
```
PS C:\Windows\system32> NET START Wazuh
```

The Wazuh service was started successfully.

```
PS C:\Windows\system32>
```



I was able to get the manager to recognize the other machine. After this was done next step was to download and configure Sysmon for better and more efficient logs.



Downloaded Sysmon for easier event logging

```
Administrator: Windows PowerShell

System Monitor v15.20 - System activity monitor
By Mark Russinovich and Thomas Garnier
Copyright (C) 2014-2026 Microsoft Corporation
Using libxml2. libxml2 is Copyright (C) 1998-2012 Daniel Veillard. All Rights Reserved.
Sysinternals - www.sysinternals.com

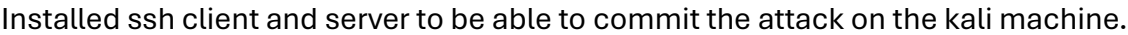
Error: Failed to open xml configuration: Downloads\sysmonconfig.xml
PS C:\Users\admin\Downloads> sysmon64.exe -c sysmonconfig.xml

System Monitor v15.20 - System activity monitor
By Mark Russinovich and Thomas Garnier
Copyright (C) 2014-2026 Microsoft Corporation
Using libxml2. libxml2 is Copyright (C) 1998-2012 Daniel Veillard. All Rights Reserved.
Sysinternals - www.sysinternals.com

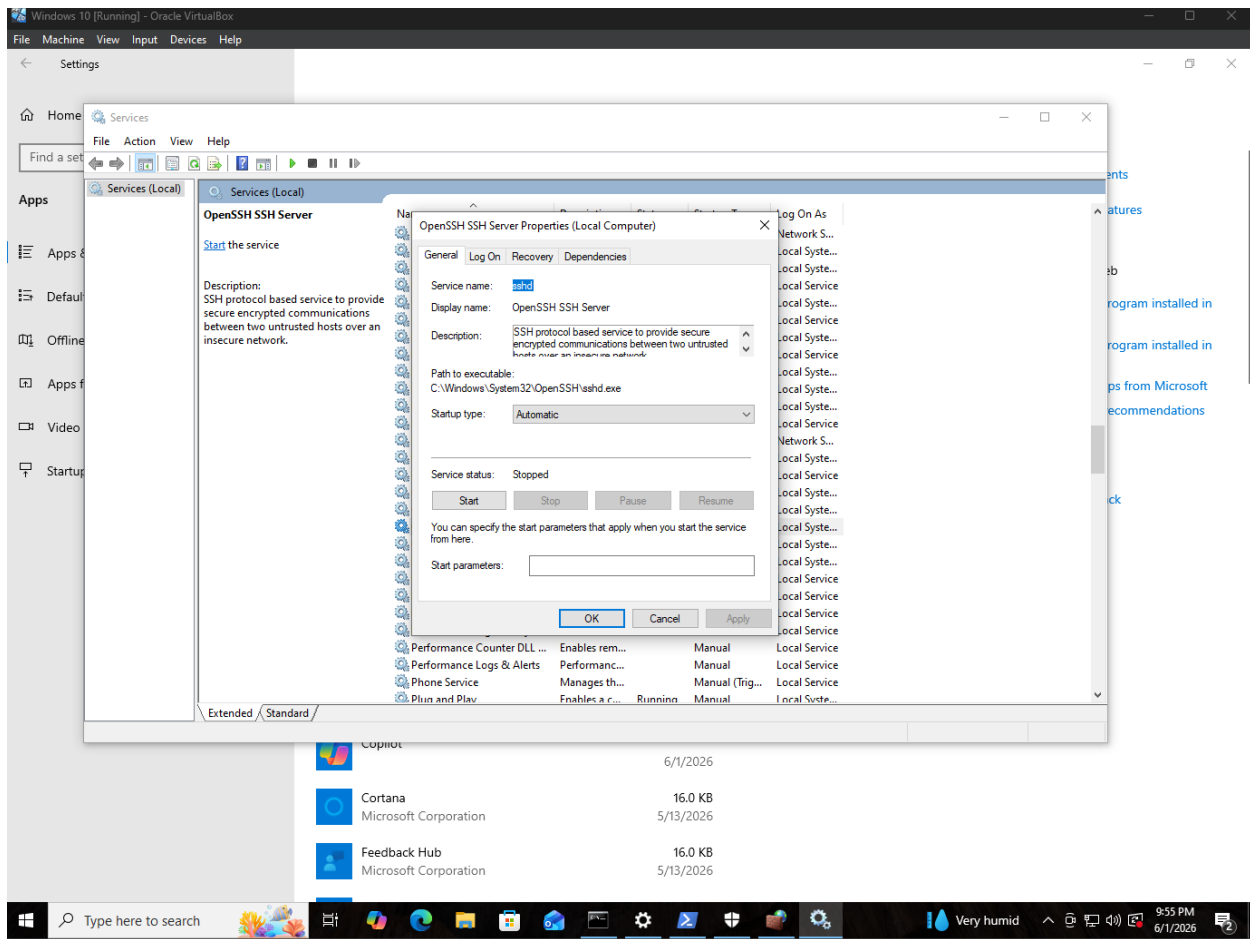
Loading configuration file with schema version 4.90
Sysmon schema version: 4.91
Configuration file validated.
Configuration updated.

PS C:\Users\admin\Downloads>
```

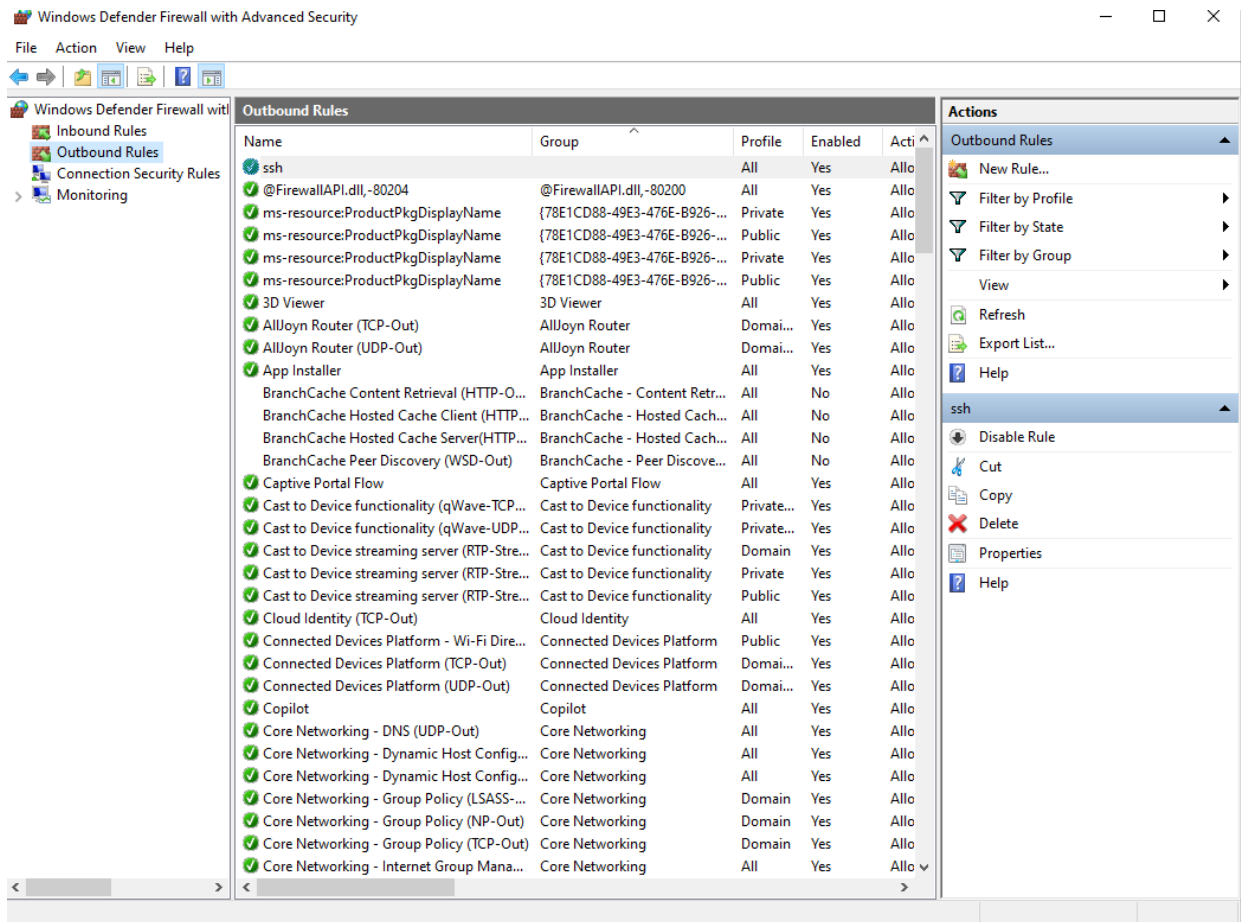
Added configuration file from GitHub repository to Sysmon to consolidate logs







Set ssh Server to automatic so it runs on startup



Then set a firewall rule to allow port 22 connections.

DQL

Last 24 hours

May 25, 2026 @ 00:00

wazuh-alerts-\*

DataOptions

Metrics

Slice size

Aggregation

Count

Custom label

Buckets

Split slices

Aggregation

Date Range

Field

timestamp

Acceptable date formats

now-1w/w

vboxuser@ubuntu: ~

GNU nano 8.7.1 /var/ossec/etc/ossec.conf \*

```
<!--
Wazuh - Manager - Default configuration for ubuntu 26.04
More info at: https://documentation.wazuh.com
Mailing list: https://groups.google.com/forum/#!forum/wazuh
-->

<ossec_config>
  <global>
    <jsonout_output>yes</jsonout_output>
    <alerts_log>yes</alerts_log>
    <logall>no</logall>
    <logall_json>yes</logall_json>
    <email_notification>no</email_notification>
    <smtp_server>smtp.example.wazuh.com</smtp_server>
    <email_from>wazuh@example.wazuh.com</email_from>
    <email_to>recipient@example.wazuh.com</email_to>
    <email_maxperhour>12</email_maxperhour>
    <email_log_source>alerts.log</email_log_source>
    <agents_disconnection_time>15m</agents_disconnection_time>
    <agents_disconnection_alert_time>0</agents_disconnection_alert_time>
  </global>
</ossec_config>
```

Help

Write Out

Where Is

Cut

Execute

Location

Exit

Read File

Replace

Paste

Justify

Go To Line

Used the command “sudo nano /var/ossec/etc/ossec.conf” to change the <logall\_json> from no to yes, so I can monitor all logs coming from the windows agent.

May 25, 2026 @ 00:00

wazuh-al

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vboxuser@ubuntu: ~

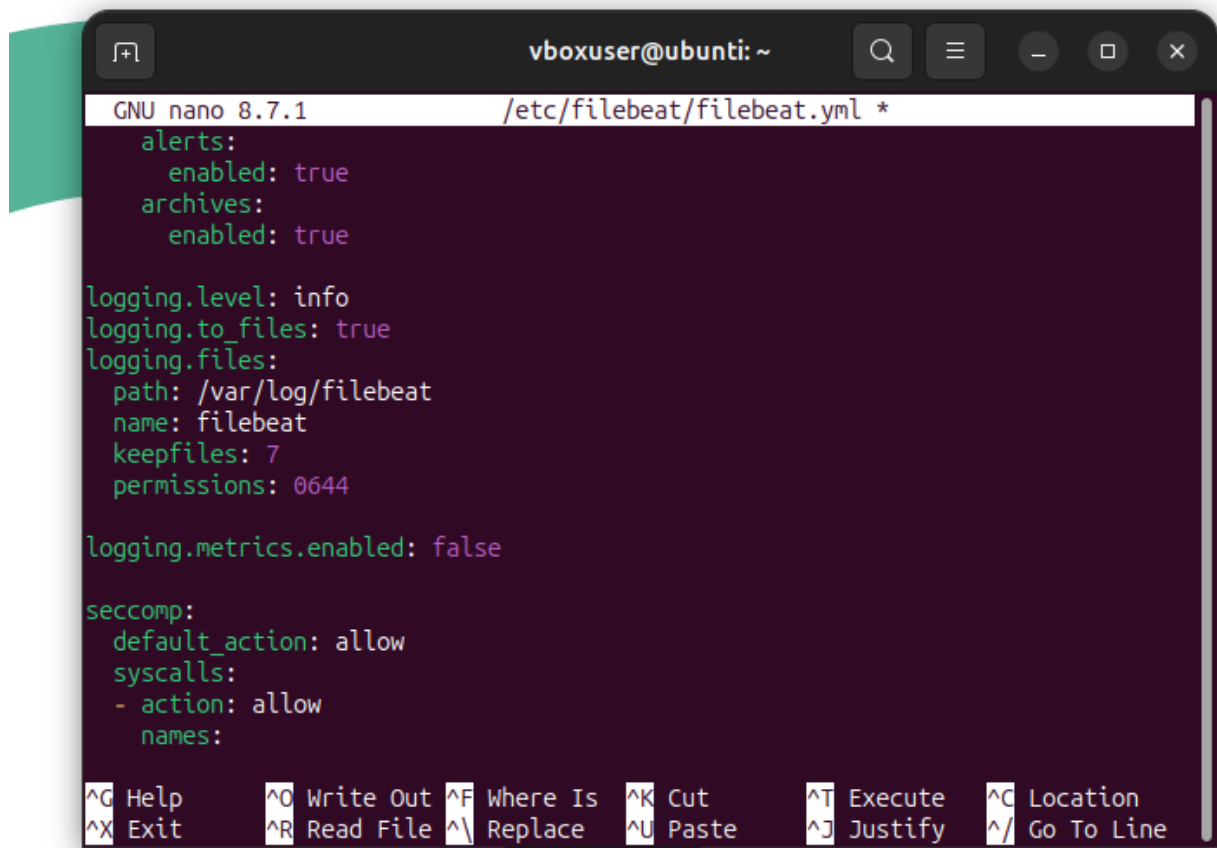
Q ≡ - □ ×

vboxuser@ubuntu:~\$ sudo nano /var/ossec/etc/ossec.conf

[sudo: authenticate] Password:

vboxuser@ubuntu:~\$ sudo systemctl restart wazuh-manager

Restarted the manager using the “sudo systemctl restart Wazuh-manager” command to finalize the changes I made in the config file.



```
GNU nano 8.7.1 /etc/filebeat/filebeat.yml *
  alerts:
    enabled: true
  archives:
    enabled: true

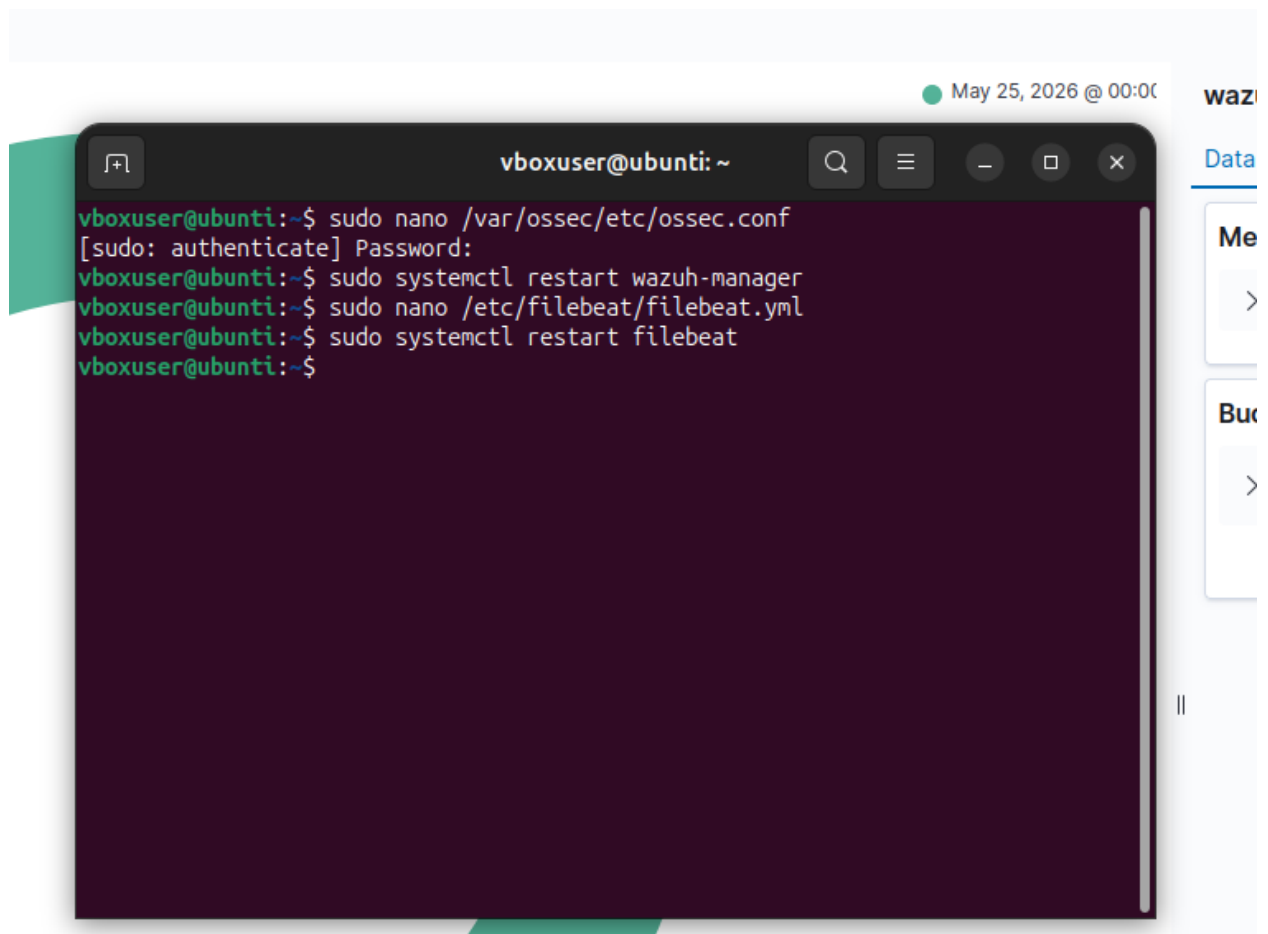
logging.level: info
logging.to_files: true
logging.files:
  path: /var/log/filebeat
  name: filebeat
  keepfiles: 7
  permissions: 0644

logging.metrics.enabled: false

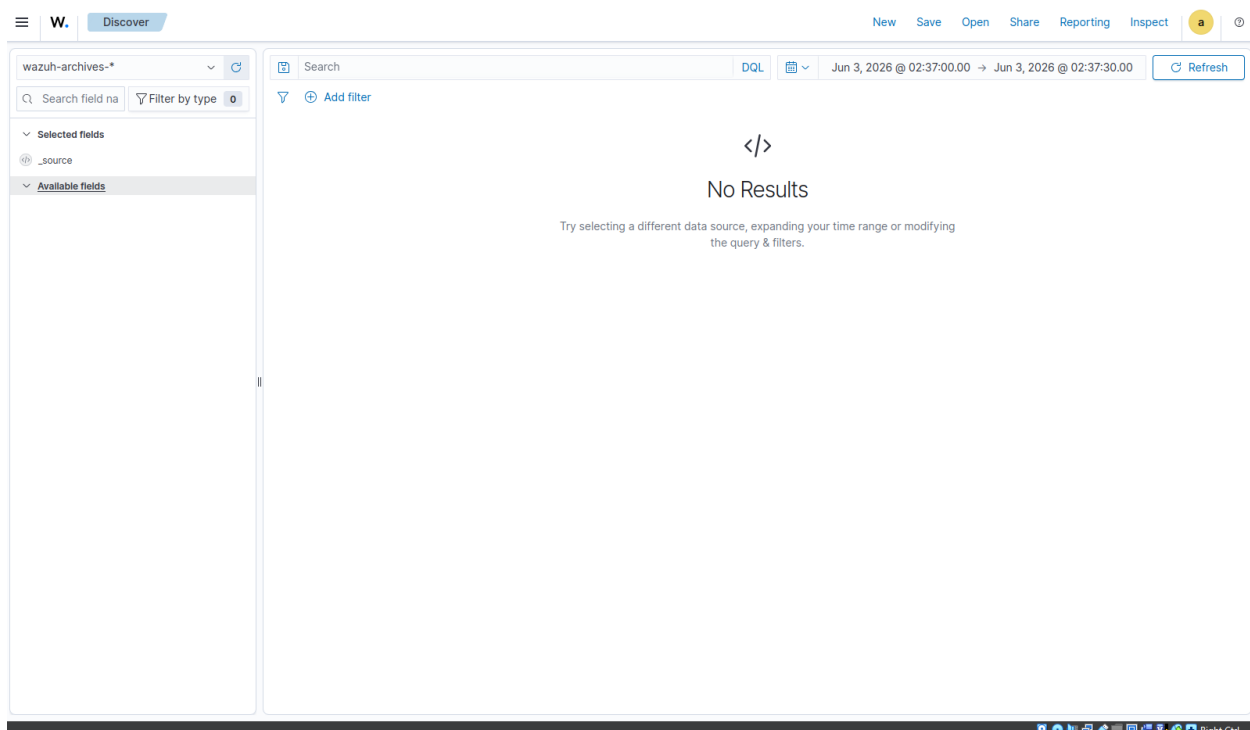
seccomp:
  default_action: allow
  syscalls:
    - action: allow
      names:
```

^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute ^C Location  
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^\_ Go To Line

Then used “sudo nano /etc/filebeat/filebeat.yml” to change the modules archives enabled from false to true. This is to ensure these archives show up on my dashboard.



we then restart the filebeat with the “sudo systemctl restart filebeat” command to ensure the changes are saved to the wazuh manager.



We now have a “wazuh-archives-\*” dashboard where we can view all the alerts.

```
matthew@kali: ~  
File Actions Edit View Help  
zsh: corrupt history file /home/matthew/.zsh_history  
matthew@kali ~  
$ hydra -l admin -P rockyou.txt ssh://10.0.2.8  
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organiz  
ations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).  
  
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-06-03 22:20:30  
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use  
-t 4  
[DATA] max 16 tasks per 1 server, overall 16 tasks, 14344398 login tries (l:1/p:14344398), ~896525 tries per tas  
k  
[DATA] attacking ssh://10.0.2.8:22/  
[ERROR] all children were disabled due too many connection errors  
0 of 1 target completed, 0 valid password found  
[INFO] Writing restore file because 2 server scans could not be completed  
[ERROR] 1 target was disabled because of too many errors  
[ERROR] 1 targets did not complete  
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-06-03 22:20:34
```

Using Hydra to try to connect to the Windows VM with the rockyou.txt but was not able to possibly due to the port not accepting connections which I will use nmap to verify if the port is open.

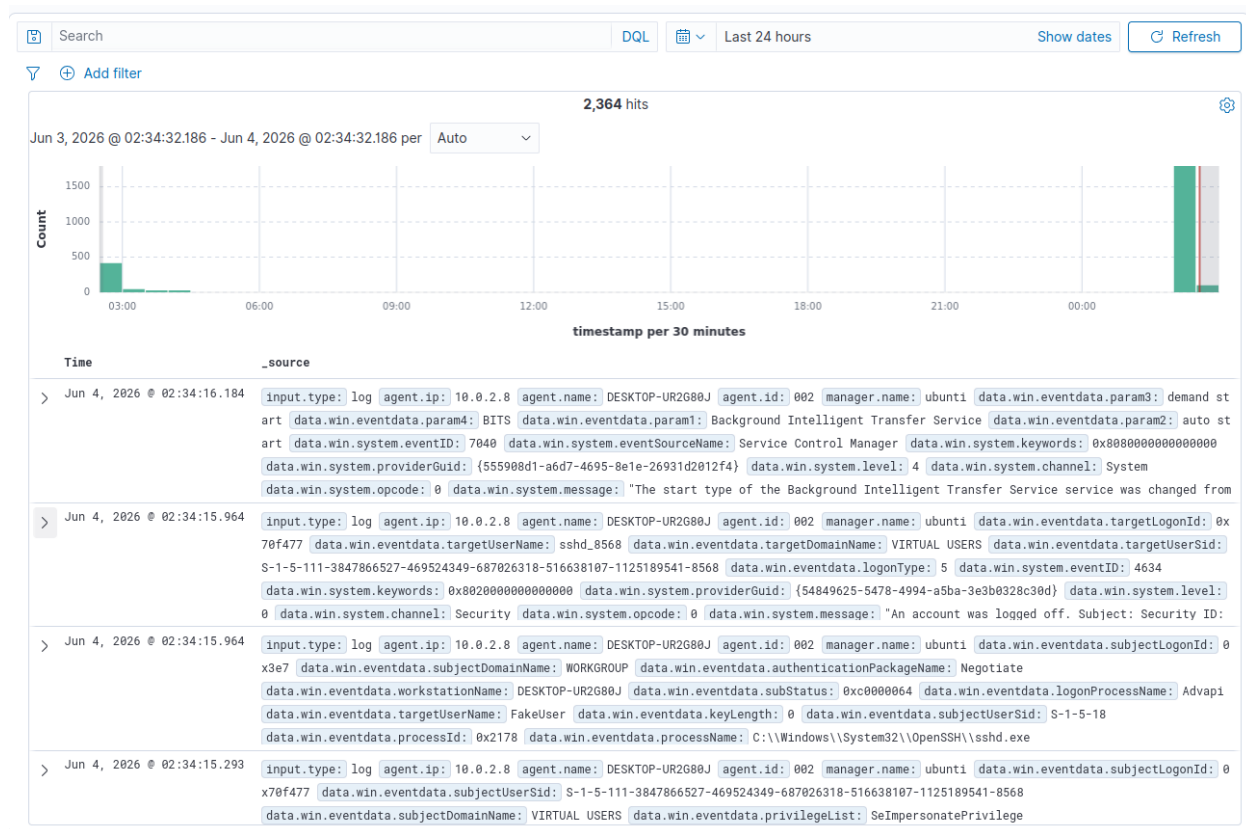
```
(matthew@kali)-[~]
$ sudo nmap -p 22 -Pn 10.0.2.8
[sudo] password for matthew:
Sorry, try again.
[sudo] password for matthew:
Starting Nmap 7.95 ( https://nmap.org ) at 2026-06-03 22:26 EDT
Nmap scan report for 10.0.2.8
Host is up (0.00023s latency).

PORT      STATE SERVICE
22/tcp    open  ssh
MAC Address: 08:00:27:B5:BB:A1 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.21 seconds

(matthew@kali)-[~]
$
```

The nmap scan using -p to specify the port and -Pn to force the scan anyway verifies the port is open.



The good news is that the logs are being reported to the SIEM dashboard



```

GNU nano 2.7.1 /var/ossec/etc/rules/local_rules.xml
group name="local,syslog,sshd,">

<!--
Dec 10 01:02:02 host sshd[1234]: Failed none for root from 1.1.1.1 port 1066 -->
-->
<rule id="100001" level="5">
  <if_sid>5716</if_sid>
  <srcip>1.1.1.1</srcip>
  <description>sshd: authentication failed from IP 1.1.1.1.</description>
  <group>authentication_failed,pci_dss_10.2.4,pci_dss_10.2.5,</group>
</rule>

<rule id="100016" level="10" frequency="5" timeframe="60">
  <if_matched_sid>60112</if_matched_sid>
  <same_source_ip />
  <description>Possible SSH Brute Force Attack: Multiple failed logins detected</description>
  <group>bruteforce,windows_auth,</group>
</rule>
</group>

```

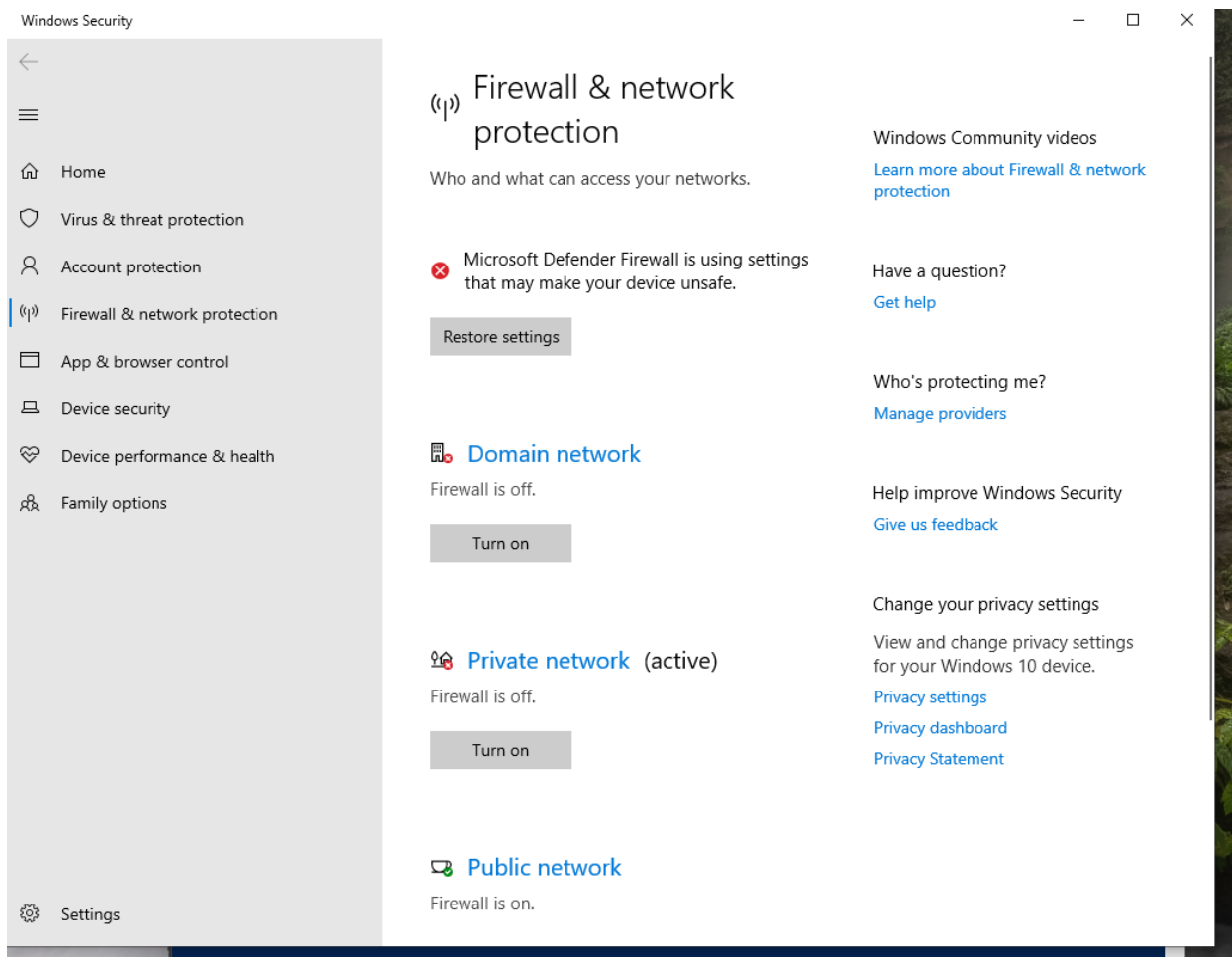
Created a rule in the Wazuh rules for brute force ssh to be alerted when an attempt happens more than 5 times.

```

vboxuser@ubuntu: ~
[sudo: authenticate] Password:
vboxuser@ubuntu:~$ sudo nano /var/ossec/etc/rules/local_rules.xml
vboxuser@ubuntu:~$ sudo /var/ossec/bin/wazuh-logtest
Starting wazuh-logtest v4.14.5
Type one log per line

```

Then using the wazuh-logtest utility then tested if the command had any errors. And then restarted the Wazuh manager for the rule to take effect.



I have also disabled the windows defender firewall to test if that is what causing my kali hydra error with the to many connections. When running an nmap scan of port 22 I am presenting with this output.

```
(matthew@kali)-[~]  
$ nc -zv 10.0.2.8 22  
10.0.2.8: inverse host lookup failed: Unknown host  
(UNKNOWN) [10.0.2.8] 22 (ssh) open
```

Which verifies that the port is open and that something else is denying the action.

```

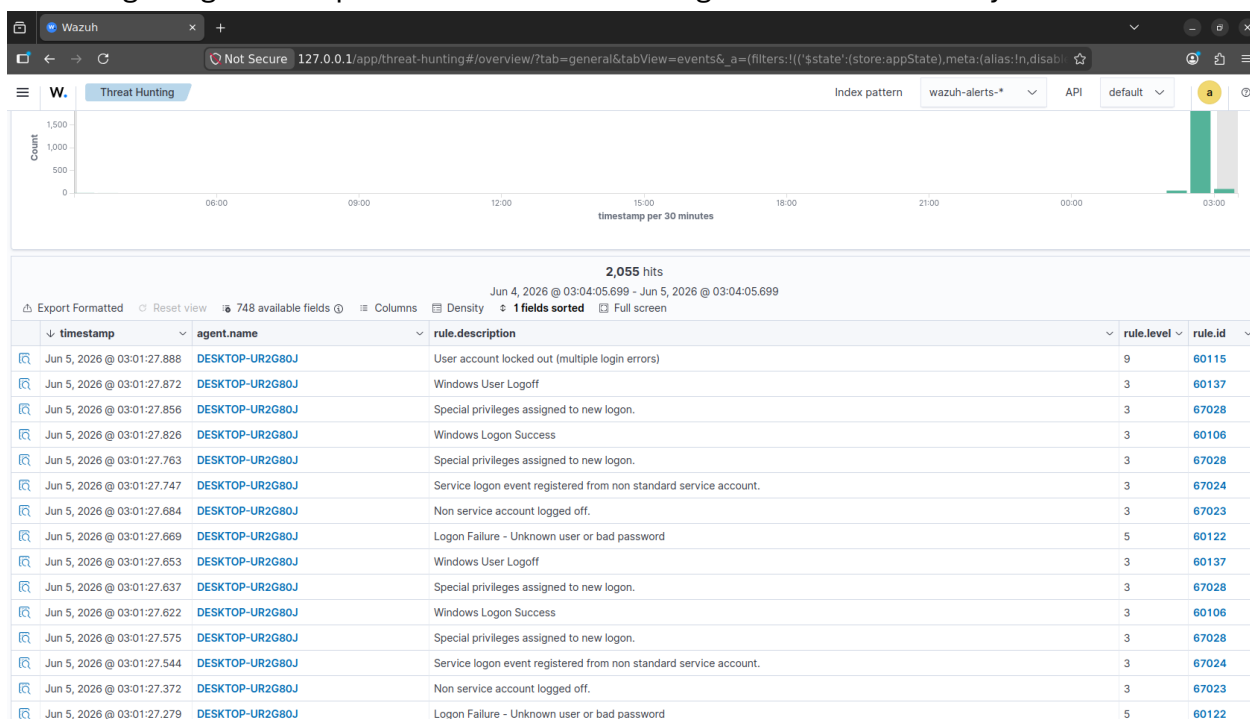
(matthew@kali)-[~]
$ hydra -l admin -P rockyou.txt -t 1 -T 1 -S -vV ssh://10.0.2.8

Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organiz
ations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

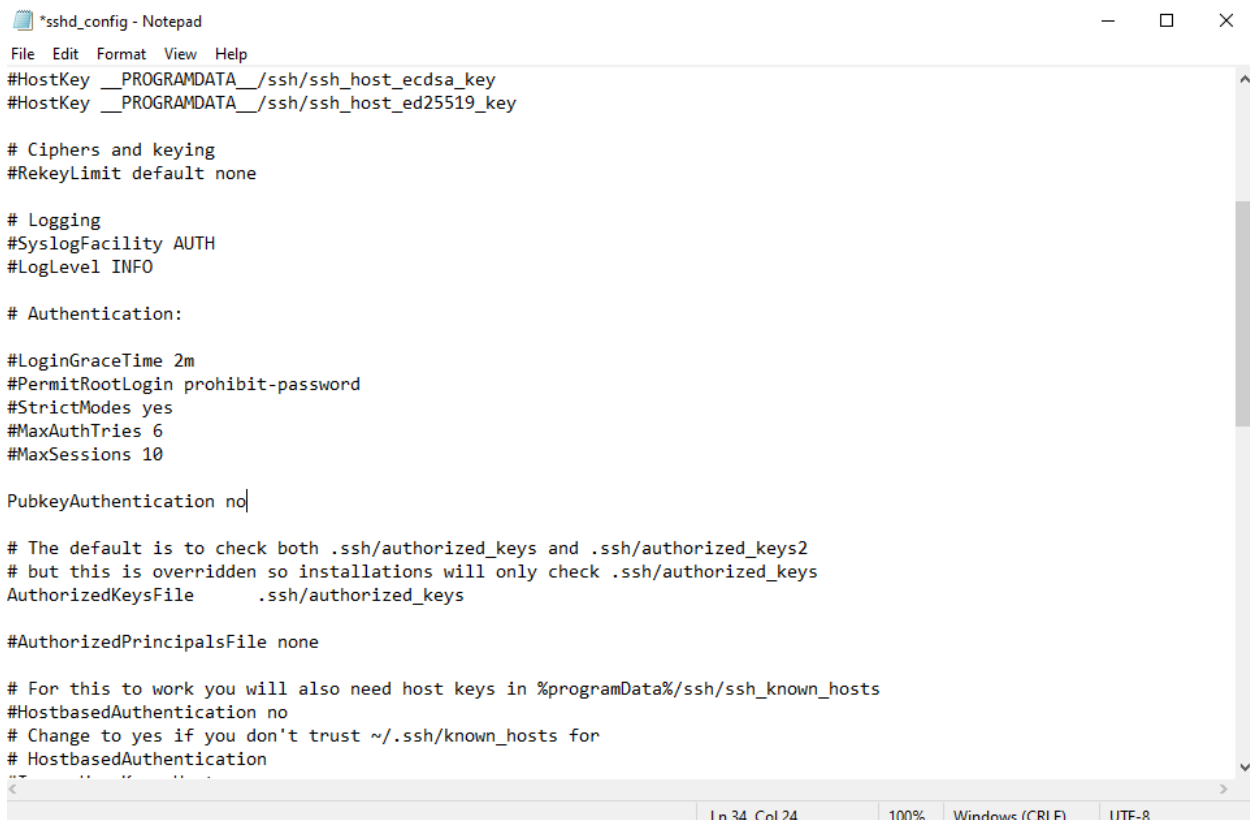
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-06-04 22:53:04
[DATA] max 1 task per 1 server, overall 1 task, 14344398 login tries (l:1/p:14344398), ~14344398 tries per task
[DATA] attacking sshs://10.0.2.8:22/
[VERBOSE] Resolving addresses ... [VERBOSE] resolving done
[INFO] Testing if password authentication is supported by ssh://admin@10.0.2.8:22
[WARNING] invalid SSH method reply from ssh://10.0.2.8:22, continuing anyway ... (check for empty password!)
[ATTEMPT] target 10.0.2.8 - login "admin" - pass "123456" - 1 of 14344398 [child 0] (0/0)
[ERROR] ssh protocol error
[ATTEMPT] target 10.0.2.8 - login "admin" - pass "12345" - 2 of 14344399 [child 0] (0/1)
[ERROR] ssh protocol error
[ATTEMPT] target 10.0.2.8 - login "admin" - pass "123456789" - 3 of 14344400 [child 0] (0/2)
[ERROR] ssh protocol error
[ERROR] all children were disabled due too many connection errors
0 of 1 target completed, 0 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-06-04 22:53:06

```

When running a Hydra command that outputs all the inputs its actively putting in, it shows that it is getting to three passwords before disabling due to the “too many connections.”



But in the Wazuh manager we do see the connections are reaching but there if a failure which is leading me to believe this is due to a public key failure which I need to figure out a way around.



```
*sshd_config - Notepad
File Edit Format View Help
#HostKey _PROGRAMDATA_/ssh/ssh_host_ecdsa_key
#HostKey _PROGRAMDATA_/ssh/ssh_host_ed25519_key

# Ciphers and keying
#RekeyLimit default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
#PermitRootLogin prohibit-password
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

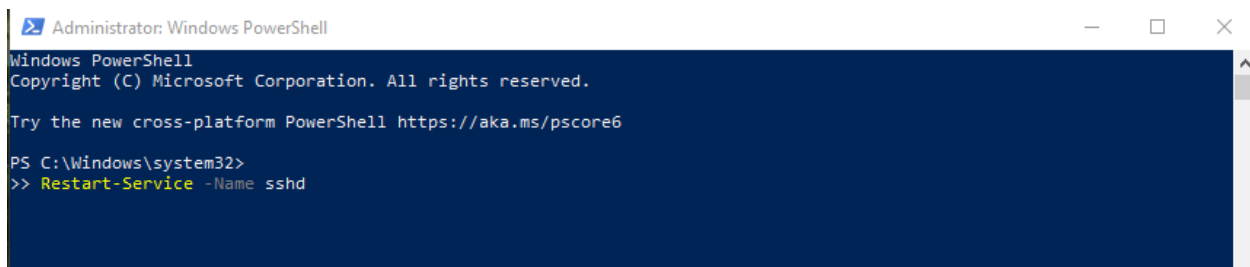
PubkeyAuthentication no

# The default is to check both .ssh/authorized_keys and .ssh/authorized_keys2
# but this is overridden so installations will only check .ssh/authorized_keys
AuthorizedKeysFile .ssh/authorized_keys

#AuthorizedPrincipalsFile none

# For this to work you will also need host keys in %programData%/ssh/ssh_known_hosts
#HostbasedAuthentication no
# Change to yes if you don't trust ~/.ssh/known_hosts for
# HostbasedAuthentication
"
```

For the sake of this lab I am turning off Public key authentication to see how the SIEM notifications work and make sure password authentication is on.

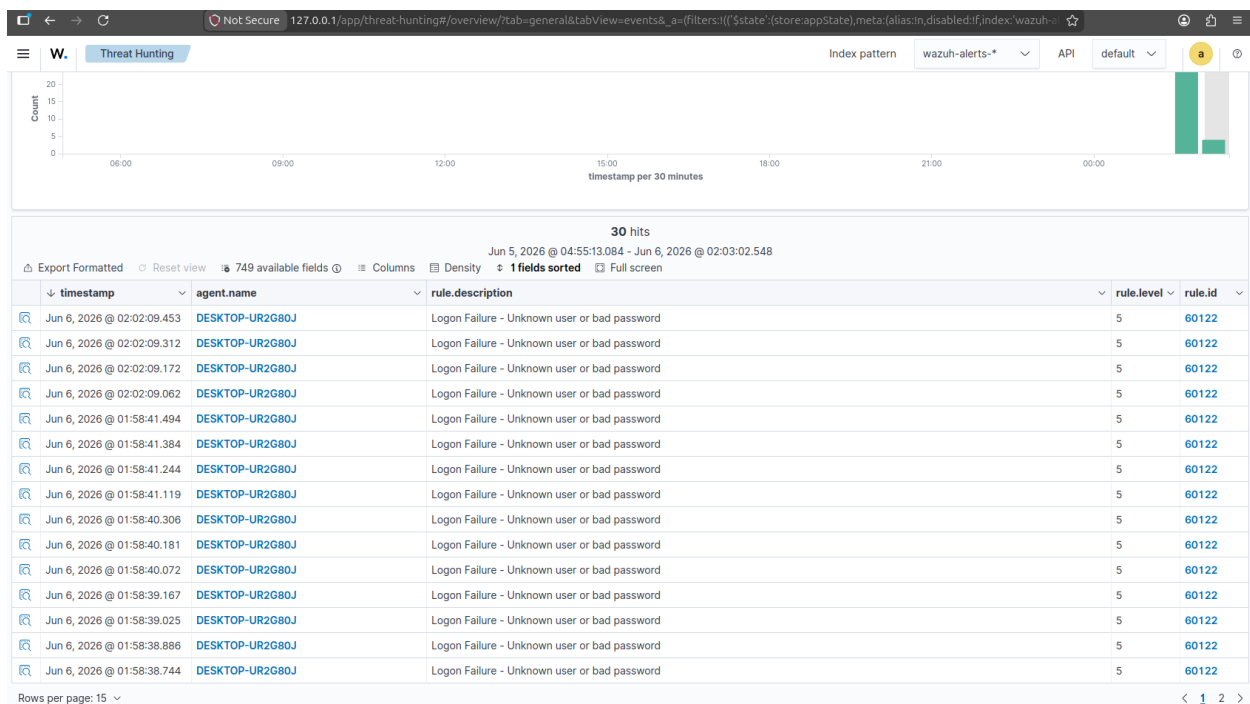


```
Administrator: Windows PowerShell
Windows PowerShell
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Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Windows\system32>
>> Restart-Service -Name sshd
```

Then we restart the ssh service.



Applying a filter in the dashboard to just show all the bad password attempts is showing in the Wazuh dashboard which was our main objective.

**Conclusion/Triage:** With the configuration of Sysmon for enhancing the logging and ensuring password authentication was enforced by disabling the public key authentication, the Wazuh manager successfully ingested resulting in authentication failures. How a SOC analyst would break this down would be first detection, with the custom rule in place which I created to alert when more than five failed SSH login attempts occur within 60 seconds. After that the next step would be the investigation phase where the SOC analyst would review the SIEM logs which would reveal a high volume of Event ID 5 or otherwise known as logon failures originating from a single IP address internally. With the rapid succession of these attempts shows a clear indication of an automated script pointing toward a compromise rather than just user error. The last step would be remediation, if this were in a real environment the first part of the remediation would be the containment of the compromise which would be blocking the offending IP address at the firewall level and verifying that there were no successful logon events, lastly you would have to reset that account password to something that is up to standard with your company's password policy.